

Reviewing Deep Learning and Convolutional Neural Networks

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UNIVERSIDAD
DE GRANADA



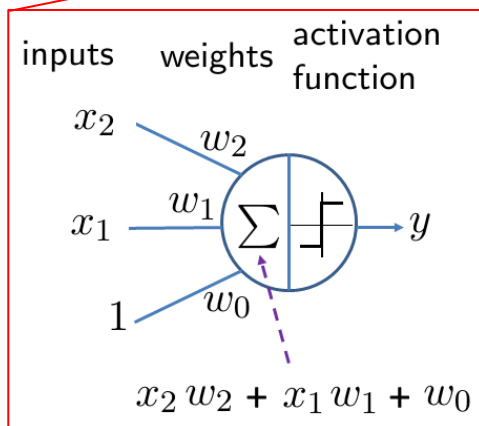
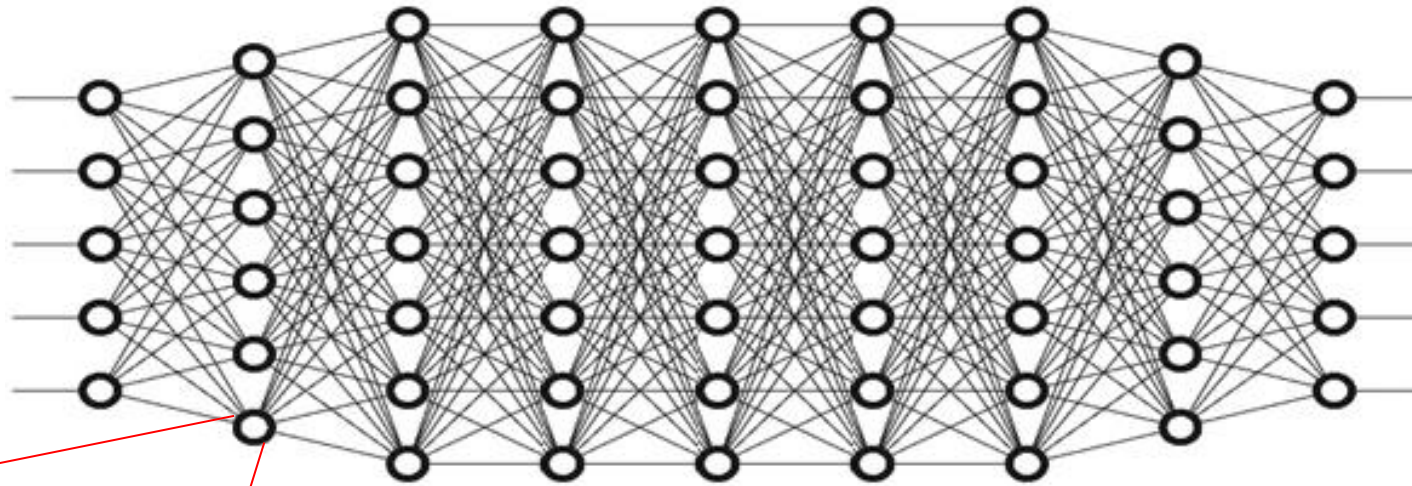
DaSCI

Instituto Andaluz de Investigación en
Data Science and Computational Intelligence

- We'll review some of the main concepts related with Convolutional Neural Networks (ConvNets) and Deep Learning (DL) using quizzes!
- Some of these questions are exam-like quizzes! Others are just to review basic concepts.

Question 1

- Could this architecture be considered a ConvNet?



Every unit in the network above corresponds to a perceptron like the one displayed on the left. Furthermore, the network also incorporates batch normalization and dropout.

Question 2

- Which layer is the main responsible for downsampling and reducing the spatial dimensions in a ConvNet?
 - A. Convolutional layer
 - B. Pooling layer
 - C. Fully connected layer
 - D. Batch normalization layer

Question 3

- In a ConvNet, what is the main purpose of the activation function typically used after the convolutional and fully connected layers?
 - A. Faster training
 - B. Introduce non-linearity
 - C. Improve regularization
 - D. Normalize input data

Question 4

- What is the main purpose of the kernel (filter) in a convolutional layer?
 - A. Reducing dimensionality
 - B. Feature extraction
 - C. Model regularization
 - D. Gradient computation

Question 5

- Which of the following statements about transfer learning in ConvNets is true?
 - A. It requires retraining the entire model from scratch.
 - B. It needs to manually set some of the filters.
 - C. It involves using pre-trained models on similar tasks.
 - D. It can only be applied to shallow networks.

Question 6

- What is the purpose of the "stride" parameter in a convolutional layer?
 - A. Control the learning rate
 - B. Define the size of the kernel
 - C. Specify the number of filters
 - D. Set the step size for sliding the kernel

Question 7

- What is the role of data augmentation in training deep neural networks?
 - A. Increase model complexity
 - B. Improve generalization
 - C. Reduce the number of parameters
 - D. Speed up training time

Question 8

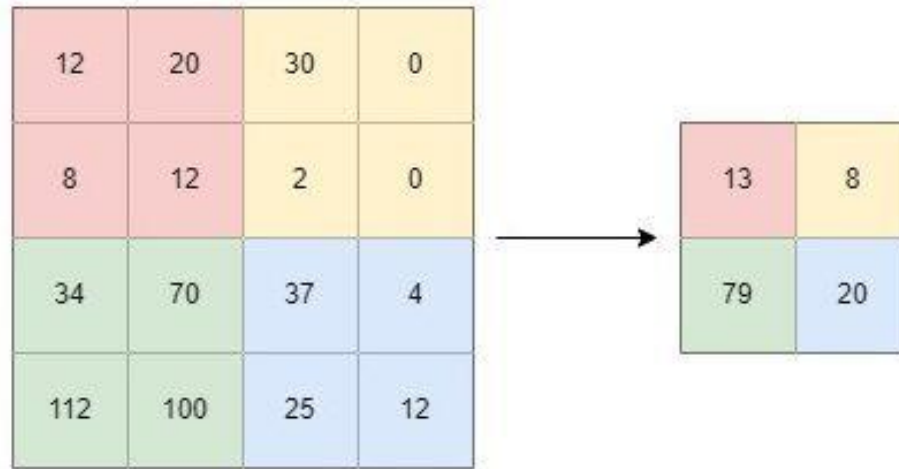
- What does the term "padding" refer to in a convolutional layer?
 - A. Adding extra layers to the model
 - B. Performing missing data imputation
 - C. Adding values (zeros, for instance) around the input volume
 - D. Normalizing the input data

Question 9

- In the context of ConvNets, what is the purpose of the softmax activation function in the output layer?
 - A. Introduce non-linearity
 - B. Normalize the output into (uncalibrated) probabilities in a regression problem
 - C. Speed up training convergence
 - D. None of the above

Question 10

- In the context of ConvNets, what kind of operation do we visualize below?



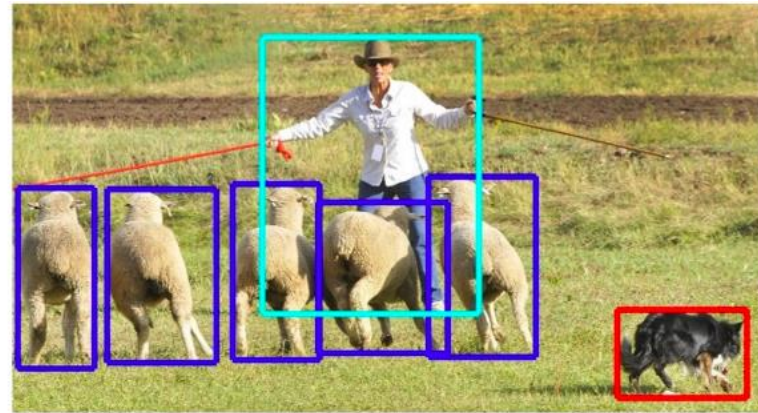
- How many extra parameters (weights) does it add to the network training?

Question 11

- Could you associate images and computer vision tasks?



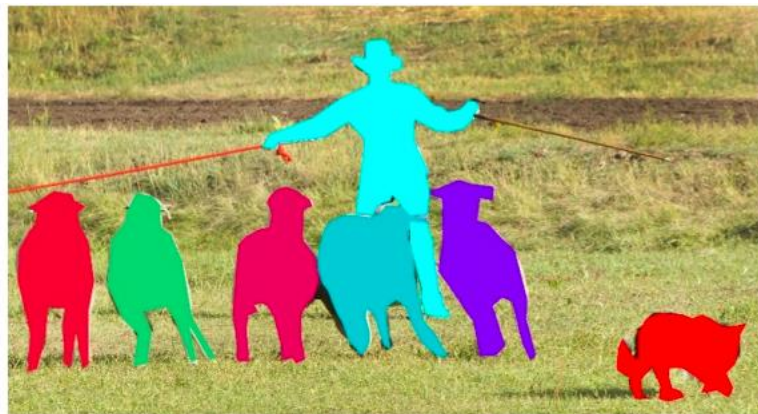
A



B



C



D

- 1) instance segmentation
- 2) object detection
- 3) image classification
- 4) semantic segmentation

Question 12

- Which of the following statements related with Dropout is incorrect?
 - A. Dropout is mainly a regularization technique
 - B. It does not introduce any extra parameter (weight)
 - C. It does not make much sense to use $p = 1$
 - D. Usually, in common frameworks, training with dropout is slower than training without it
 - E. It can only be used with fully connected layers

Question 12

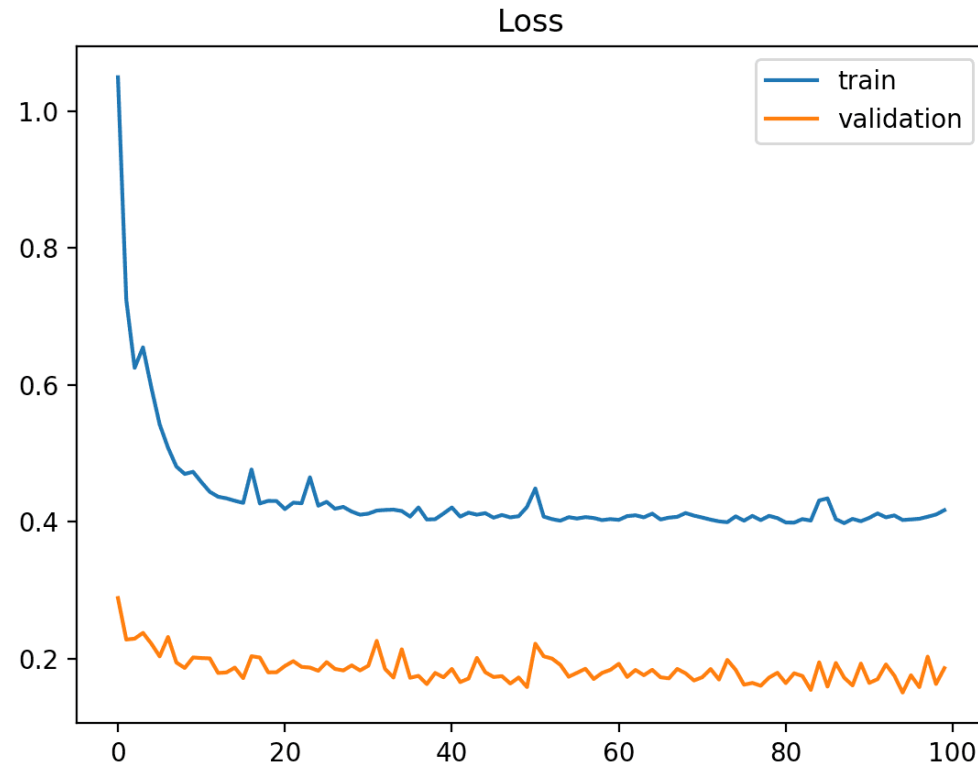
Usually, in common frameworks, **training with dropout is slower than training without it**

Generally, DL frameworks use vectorized operations on dense vectors → **the number of zeros makes no difference, since we calculate matrix operations using all entries**. It usually takes more time because we need:

- additional matrices for dropout masks
- drawing random numbers for each entry of these matrices
- multiplying the masks and corresponding activations/weights, and applying inverted dropout

Question 13

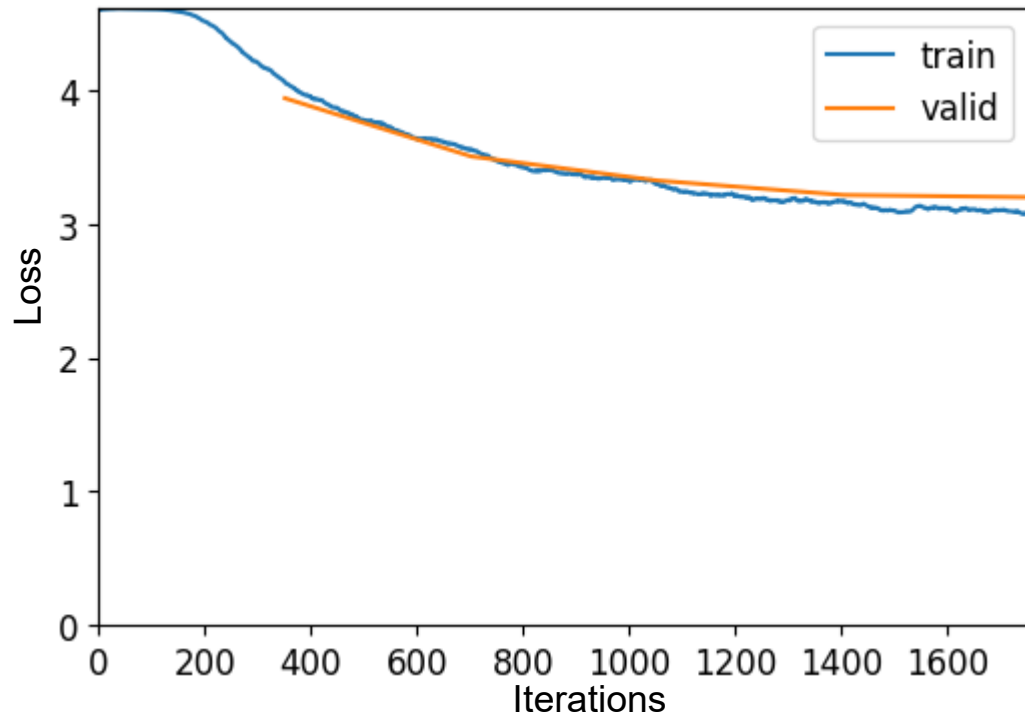
- What would be your diagnosis/hypothesis if you get the following learning curves?



- A. We are overfitting the training set. Therefore, we need regularization.
- B. Slow start: improper initialization of weights.
- C. Underfitting scenario (our model is maybe too simple, and it cannot learn almost anything from the training set).
- D. Inappropriate/Unrepresentative validation set (it seems to be much easier than the training set)

Question 14

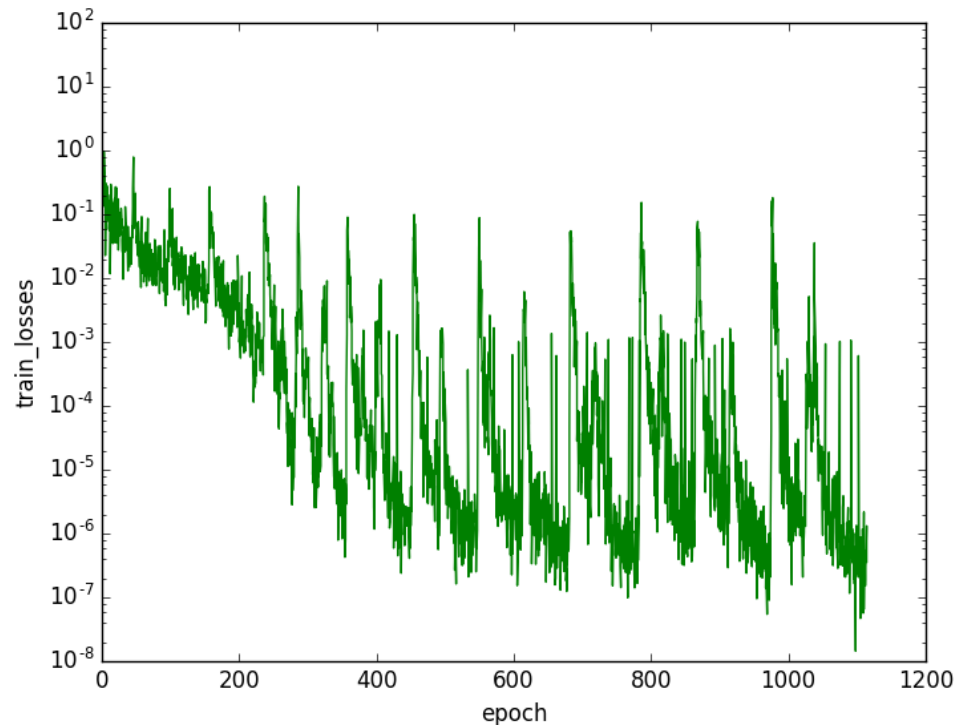
- What would be your diagnosis/hypothesis if you get the following learning curves?



- A. Overfit: model too large/dataset too small.
- B. Underfitting scenario (our model may require further training and/or higher capacity).
- C. Learning rate maybe too large.
- D. These learning curves show directly a good fit. No more training is required.

Question 15

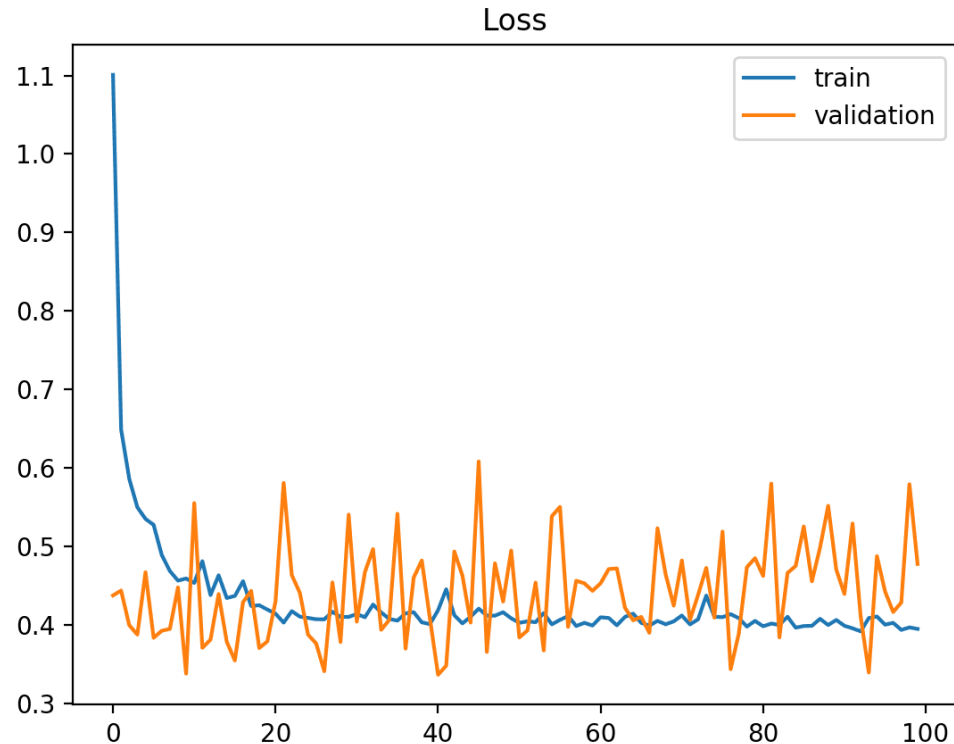
- What would be your diagnosis/hypothesis if you get the following training curve?



- A. Gradients not applied to weights.
- B. Applied the negative of gradients.
- C. Learning rate maybe too small.
- D. Not shuffling data after every epoch (periodical patterns in loss curve) or maybe learning rate too large.

Question 16

- What would be your diagnosis/hypothesis if you get the following learning curves?



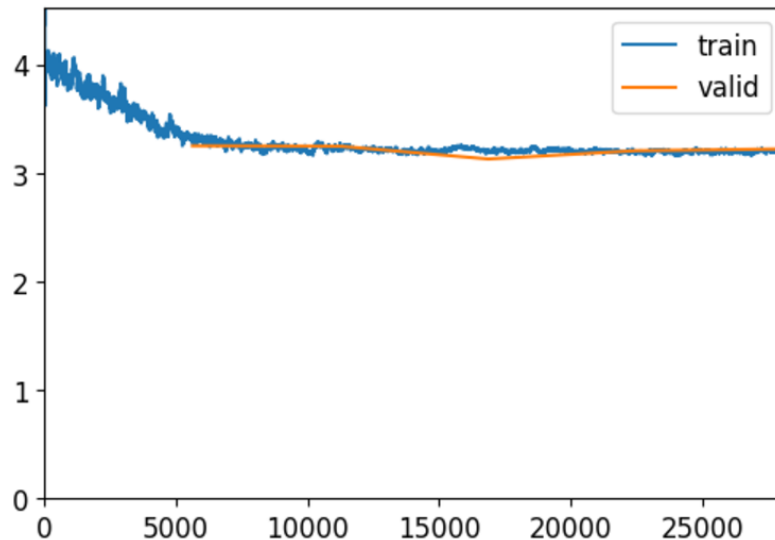
- A. Overfitting scenario: there is a large generalization gap between training and validation.
- B. Inappropriate/Unrepresentative validation set.
- C. Underfitting scenario (our model is maybe too simple, and it cannot learn the training set).
- D. Learning rate maybe too large.

Question 17

- Here, we use the exact same architecture and training strategy. The only difference is that one of the following trainings was performed with $bs = 2$ and the other with $bs = 128$. Can you associate trainings and batch size?

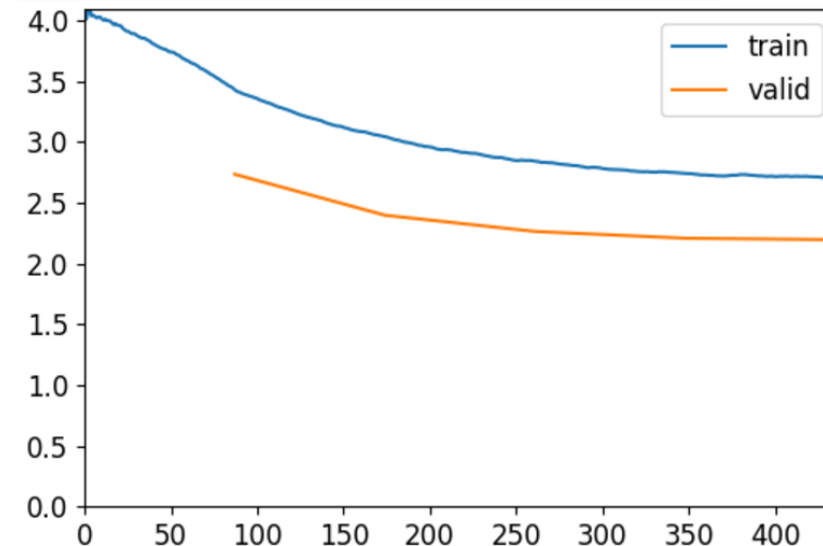
(a)

epoch	train_loss	valid_loss	accuracy	f1_score	time
0	3.317010	3.250682	0.060000	0.038292	03:40
1	3.261643	3.247044	0.041600	0.023071	03:38
2	3.205572	3.129067	0.103200	0.056297	03:41
3	3.180603	3.205932	0.067200	0.035356	03:40
4	3.194121	3.221872	0.075200	0.037645	03:39



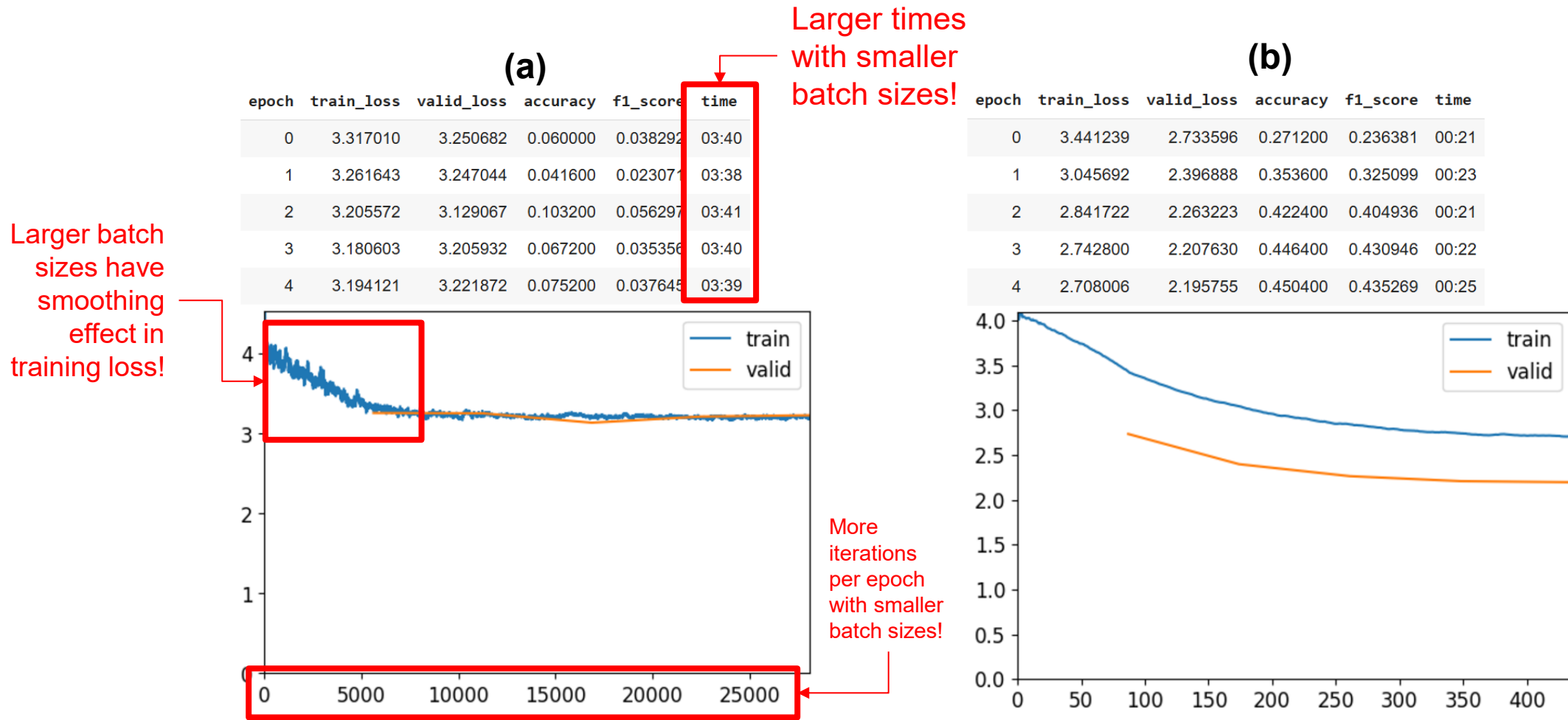
(b)

epoch	train_loss	valid_loss	accuracy	f1_score	time
0	3.441239	2.733596	0.271200	0.236381	00:21
1	3.045692	2.396888	0.353600	0.325099	00:23
2	2.841722	2.263223	0.422400	0.404936	00:21
3	2.742800	2.207630	0.446400	0.430946	00:22
4	2.708006	2.195755	0.450400	0.435269	00:25



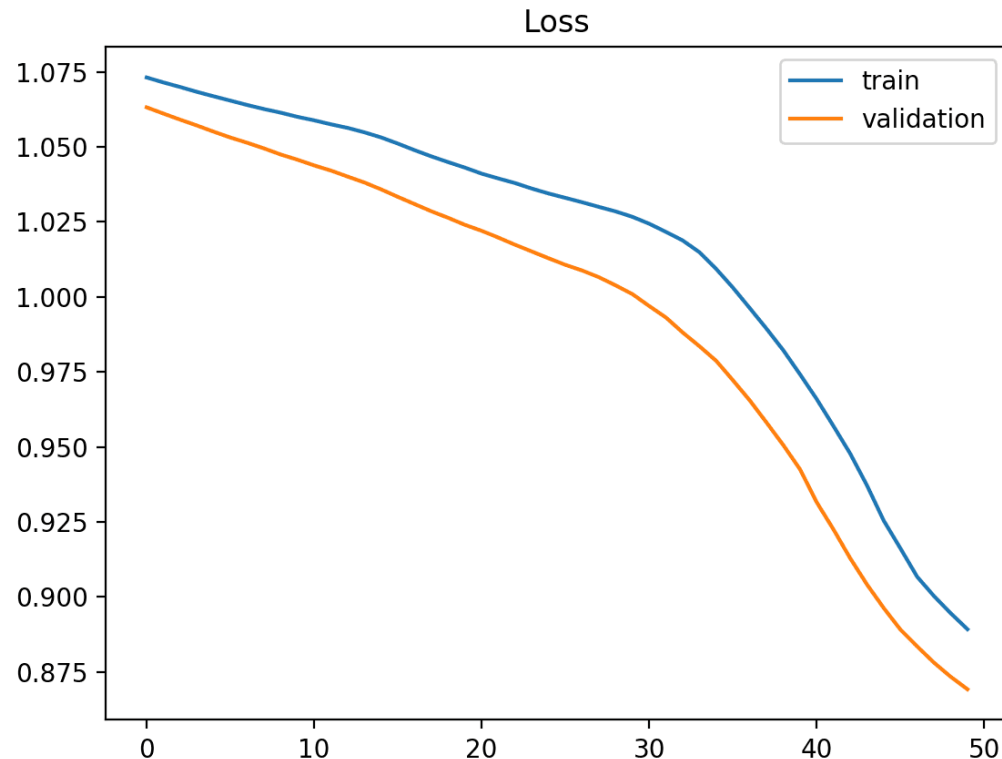
Question 17

- Here, we use the exact same architecture and training strategy. The only difference is that one of the following trainings was performed with $bs = 2$ and the other with $bs = 128$. Can you associate trainings and batch size?



Question 18

- If we are doing better in validation than in training, like in the graph below, what are some of the main reasons for that? Mark all that apply.



- A. Maybe the learning rate is too high.
- B. This is a usual situation with Adam. We may change the optimizer from Adam to AdamW.
- C. Maybe we are training with dropout (which sacrifices training accuracy to get robustness and generalization in validation).
- D. Maybe our data augmentation strategy makes training harder than validation.

Question 19

- If we have a dataset with 10.000 images and we decide to use minibatches of 8 images, how many iterations will take to complete an epoch?

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10.000 images / 8 = 1250 iterations (weight updates per epoch)

Question 20

- Let's assume the following input volume and filters used.
What is the resulting output volume?
 - Input volume: $224 \times 224 \times 3$
 - Filters used: 68 filters of 3×3 with stride 2 and no padding

Question 20

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What is the resulting output volume?
 - Input volume: 224x224x3
 - Filters used: 68 filters of 3x3 with stride 2 and no padding

$$\text{Output_size} = ((N+2*P-F)/\text{stride}) + 1$$

$$(224+2*0-3)/2 + 1 = 110.5 + 1 = 111.5 \rightarrow \text{np.floor}(111.5) = 111$$

111x111x68

Question 21

- What is the number of parameters (weights) for this convolutional architecture?

Layer Type	Kernel Size (for convolutional layers)	Input Output dimension	Input Output channels (for convolutional layers)
Conv	3x3	32x32 30x30	3 15
BN	-	30x30 30x30	-
ReLU	-	30x30 30x30	-
MaxPooling	2x2	30x30 15x15	-
Conv	3x3	15x15 13x13	15 10
BN	-	13x13 13x13	-
ReLU	-	13x13 13x13	-
Dropout	-	13x13 13x13	-
FC	-	1690 50	-
BN	-	50 50	-
ReLU	-	50 50	-
FC	-	50 25	-

Question 21

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Dropout	-	13x13 13x13	-
FC	-	1690 50	-
BN	-	50 50	-
ReLU	-	50 50	-
FC	-	50 25	-

87755

=

Weights

$(3*3*3 + 1)*15$

$2*15$

0

0

$(3*3*15 + 1)*10$

$2*10$

0

0

$1690*50 + 50$

$50*2$

0

$50*25 + 25$

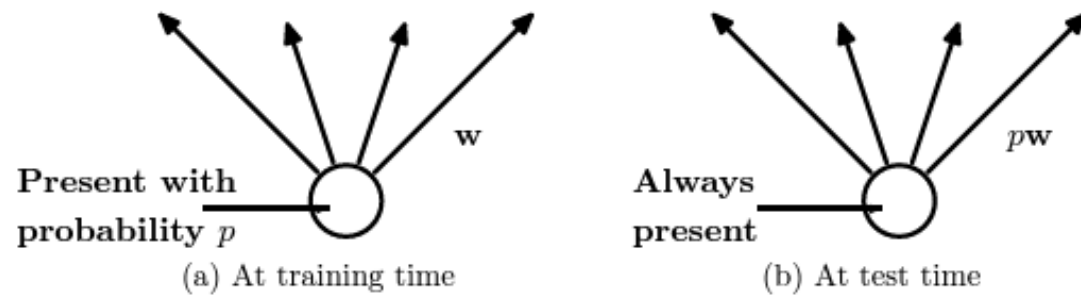
Question 22

- According to the original paper, how does Dropout (with probability p of keeping the unit) operate at inference time (validation and test)?
 - A. The same as training, removing some units.
 - B. At test time, the unit is always present and the activations are multiplied by p .
 - C. At test time, the unit is always present but you don't do anything in particular.
 - D. At test time, the unit is always present and the activations are divided by p .

Question 22

- According to the original paper, how does Dropout (with probability p of keeping the unit) operate at inference time (validation and test)?

B. At test time, the unit is always present and the weights are multiplied by p .



During inference, all units are considered. Therefore, the final weights/activations will be larger than expected, so weights are scaled by the dropout rate.

Figure 2: **Left:** A unit at training time that is present with probability p and is connected to units in the next layer with weights w . **Right:** At test time, the unit is always present and the weights are multiplied by p . The output at test time is same as the expected output at training time.

Question 23

- If you take a ConvNet and remove all non-linearities, can you improve its overall capacity (i.e., its ability to approximate more complex and non-linear functions) if you...?
 - A. Increase the number of filters/kernels per convolutional block
 - B. Increase the number of layers in your network
 - C. Increase filters/kernels sizes
 - D. Train for more epochs
 - E. None of the above

Question 23

- If you take a ConvNet and remove all non-linearities, can you improve its overall capacity (i.e., its ability to approximate more complex and non-linear functions) if you...?
 - This non-linearity is **necessary to learn complex representations of the input data.**
 - Otherwise, if only linear activation functions were used, the neural network would behave like a linear function.

If the activation functions of all the hidden units in a network are taken to be linear, then for any such network we can always find an equivalent network without hidden units.

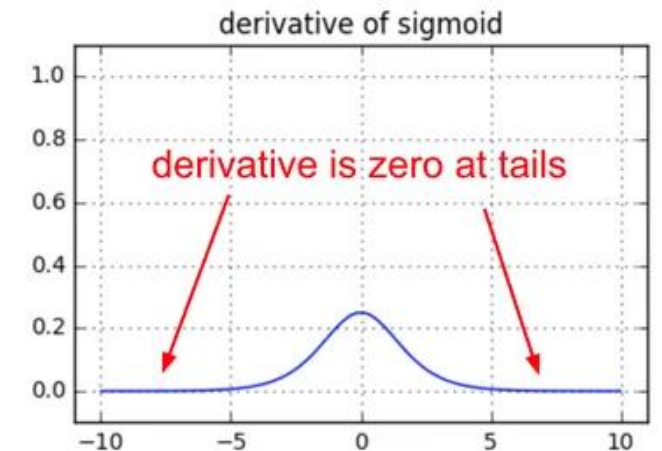
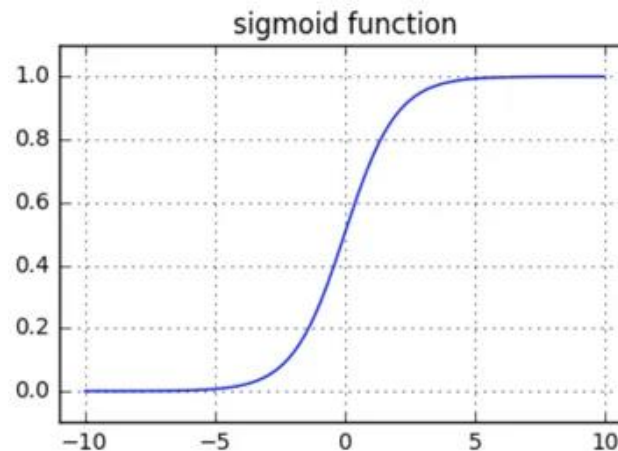
Question 24

- What is a characteristic issue associated with saturating sigmoid activation functions?
 - A. Exploding gradients
 - B. Dead neurons
 - C. Vanishing gradients
 - D. Dead ReLU units

Question 24

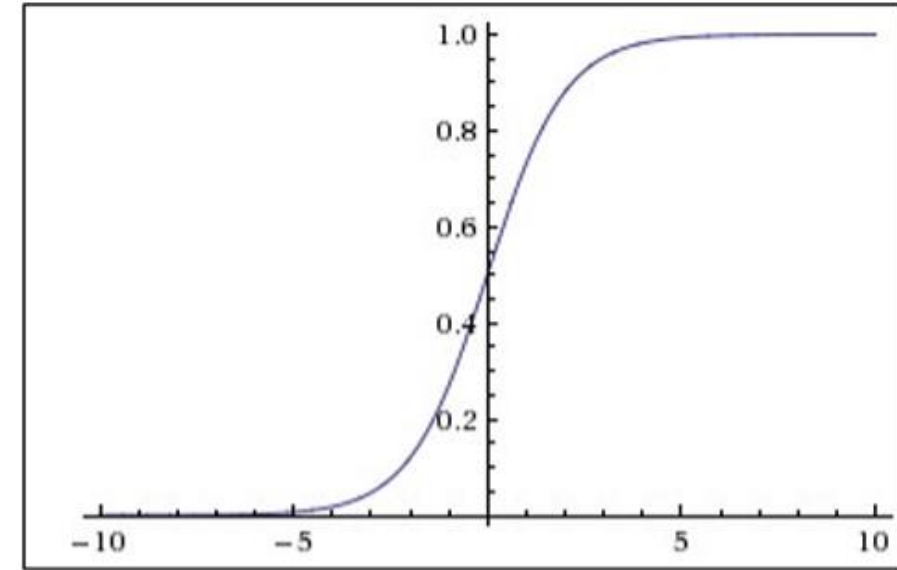
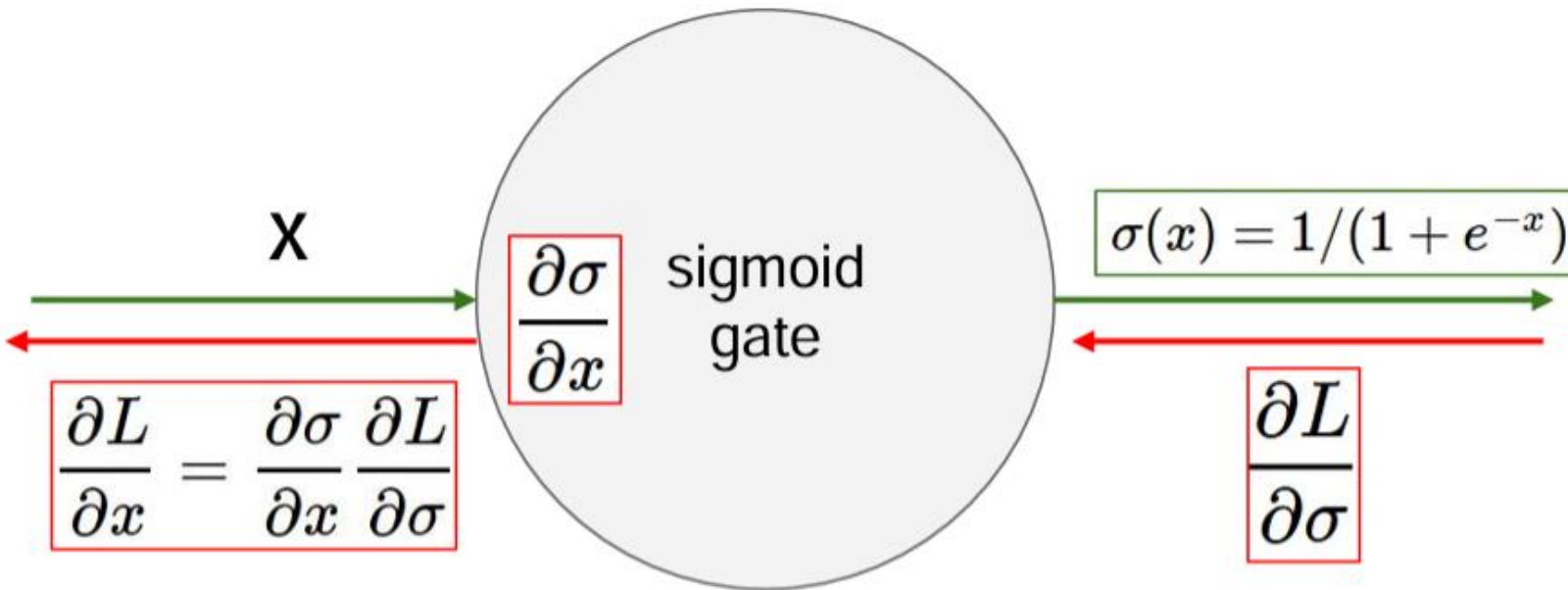
- What is a characteristic issue associated with saturating sigmoid activation functions?

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- B. Dead neurons
- C. **Vanishing gradients**
- D. Dead ReLU units



See dying ReLU and vanishing gradient problem
([“Yes, you should understand backprop”](#) by Andrej Karpathy)

Vanishing Gradient Problem

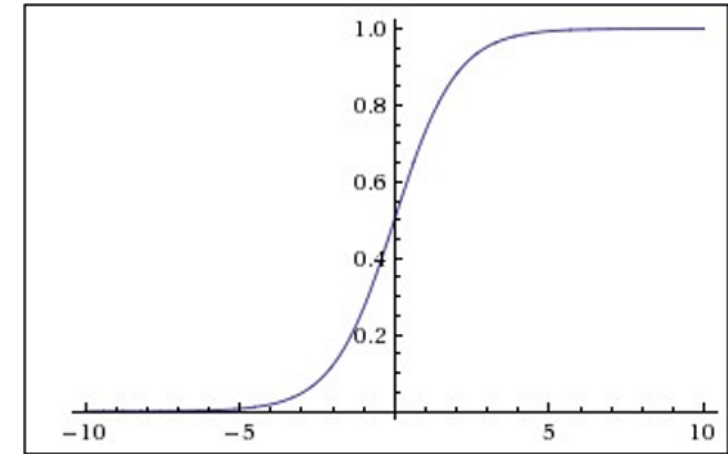
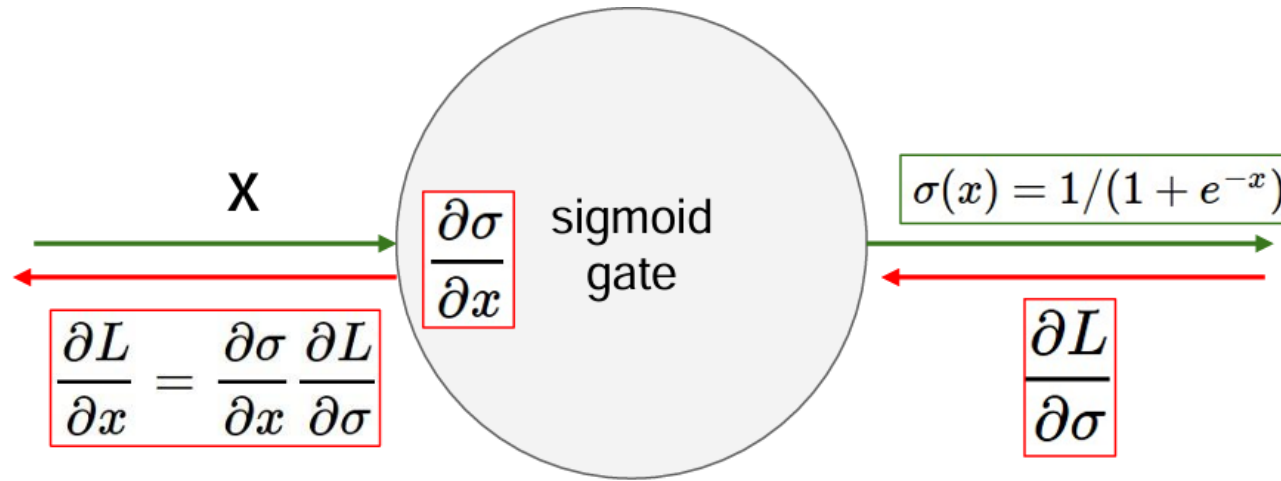


$$\frac{\partial \sigma(x)}{\partial x} = \sigma(x) (1 - \sigma(x))$$

Recommended readings:

- Intro to backprop (using computational graphs): https://cs231n.stanford.edu/slides/2024/lecture_4.pdf (slides 57-102)
- Derivation of backprop: <https://en.wikipedia.org/wiki/Backpropagation#Derivation> (quite clear and nice explanation)
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Vanishing Gradient Problem



What happens when $x = -10$?

$$\sigma(x) \approx 0$$

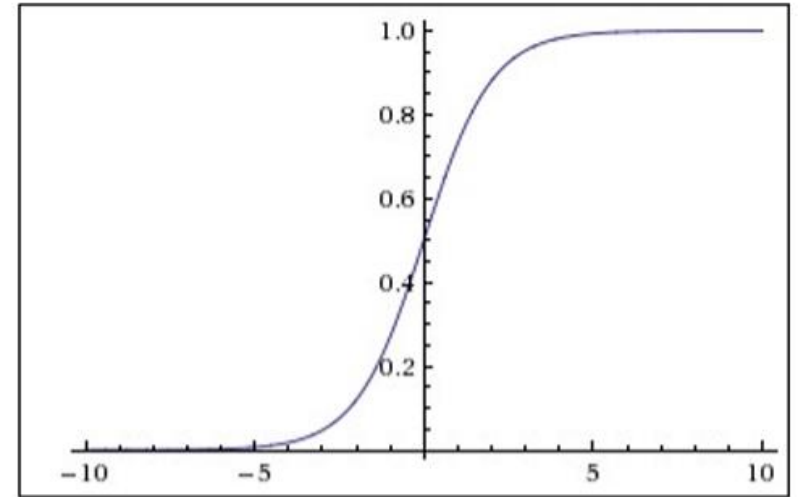
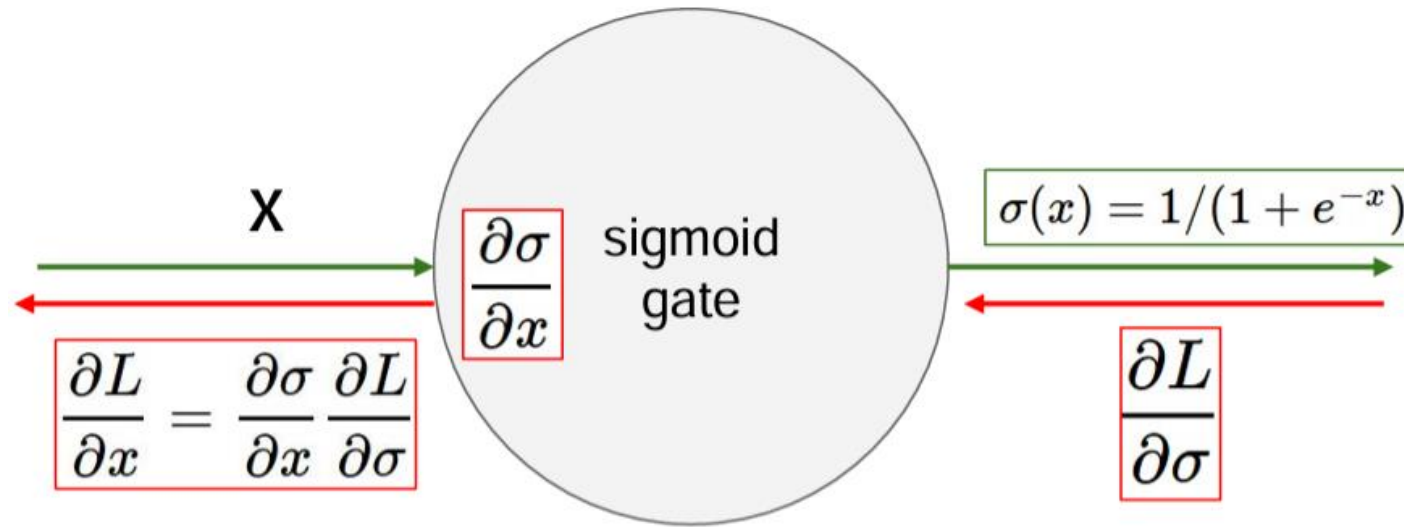
$$\frac{\partial \sigma(x)}{\partial x} = \sigma(x) (1 - \sigma(x)) = 0(1 - 0) = 0$$

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Vanishing Gradient Problem



What happens when $x = 10$?

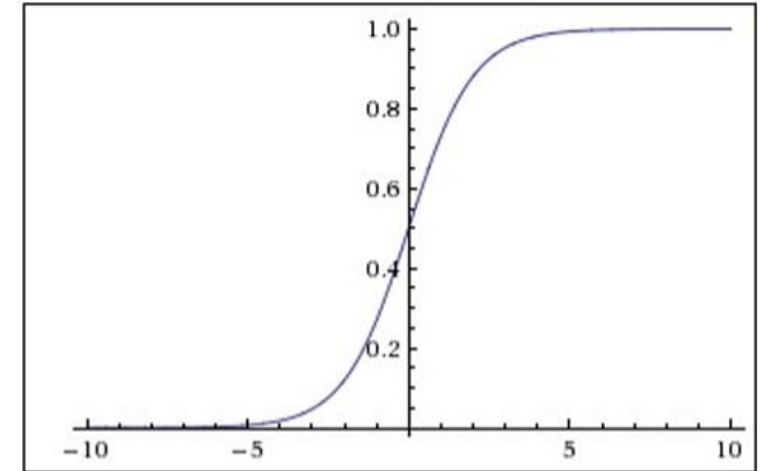
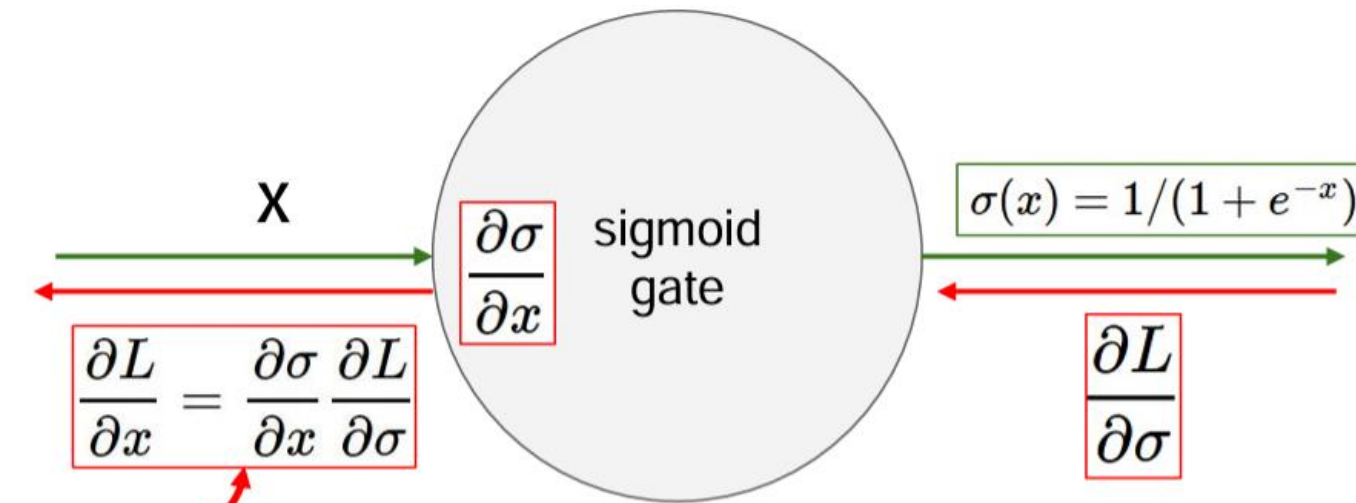
$$\frac{\partial \sigma(x)}{\partial x} = \sigma(x) (1 - \sigma(x))$$

$$\sigma(x) \approx 1 \quad \frac{\partial \sigma(x)}{\partial x} = \sigma(x) (1 - \sigma(x)) = 1(1 - 1) = 0$$

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Vanishing Gradient Problem



Why is this a problem?

$$\frac{\partial \sigma(x)}{\partial x} = \sigma(x) (1 - \sigma(x))$$

If gradients flowing back are $\sim 0 \rightarrow$ weights won't change!

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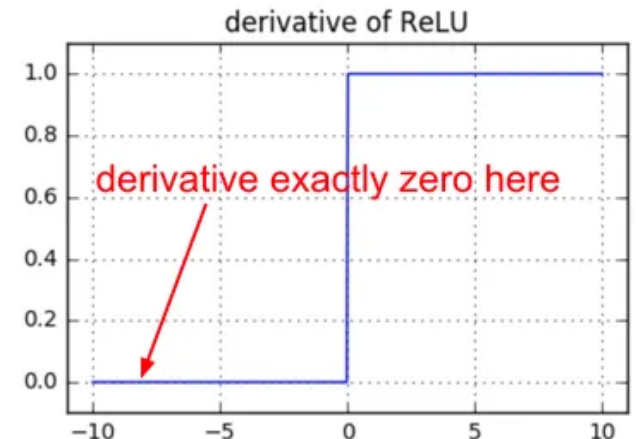
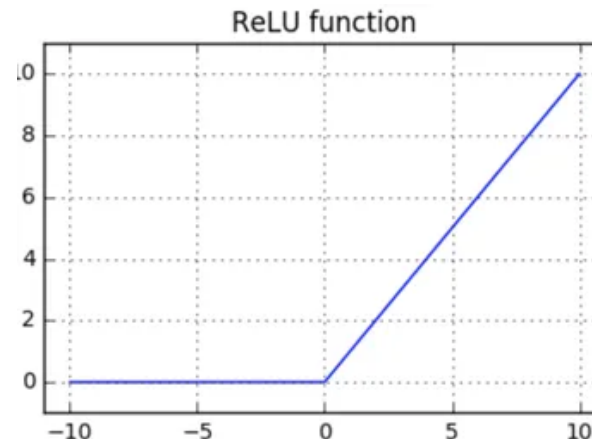
Question 25

- What is the primary effect and disadvantage of dead ReLU units in a neural network?
 - A. They lead to exploding gradients.
 - B. They contribute to overfitting.
 - C. They decrease the number of active units contributing to the learning process.
 - D. The network takes much longer to train (per epoch).

Question 25

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 - C. They decrease the number of active units contributing to the learning process.**
 - D. The network takes much longer to train (per epoch).

See dying ReLU problem and vanishing gradient problem ([“Yes, you should understand backprop”](#) by Andrej Karpathy)



Question 26

- Evaluate whether the following statements are true or false.
 - A. You should not use a linear activation function in an output unit for a multiclass classification problem. True/False
 - B. You should not use cross-entropy loss in a regression problem. True/False
 - C. A convolutional layer is equivariant to translations. True/False
 - D. Pooling helps to make the representation approximately invariant to small translations of the input. True/False

Question 27

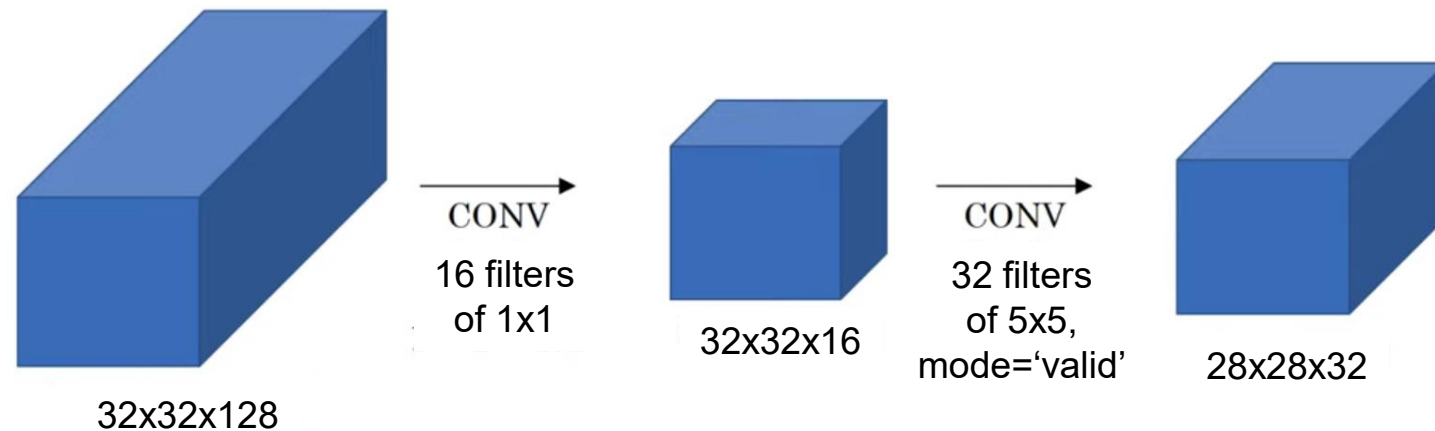
- Let's say that you found an awesome network on some related work, but the input is not the same as the one of your problem (different dimensionality)... 😞
 - A. Bad luck. Forget about it, you just cannot do anything.
 - B. Nice! I can apply it directly to my problem!
 - C. Wait! I just need to do some finetuning without adapting or modifying anything else!
 - D. Need to deal with larger/smaller input. E.g., you can stack one more conv+pool block, or change kernel size/stride

Question 28

- Let's imagine that I decide to incorporate a new type of layer into my neural model. Which of the following seems more reasonable to you?
 - A. Great! Let's include it straight away, and directly train our model!
 - B. Wait... is it differentiable? Because otherwise we cannot include it... (unless we use non-gradient-based approaches to train)
 - C. Wait... are we sure that it does not incorporate more weights to train?
 - D. It doesn't make sense to introduce more types of layers. We know from theory that convolutional layers, pooling layers, and fully connected layers are all we need to approximate any function.

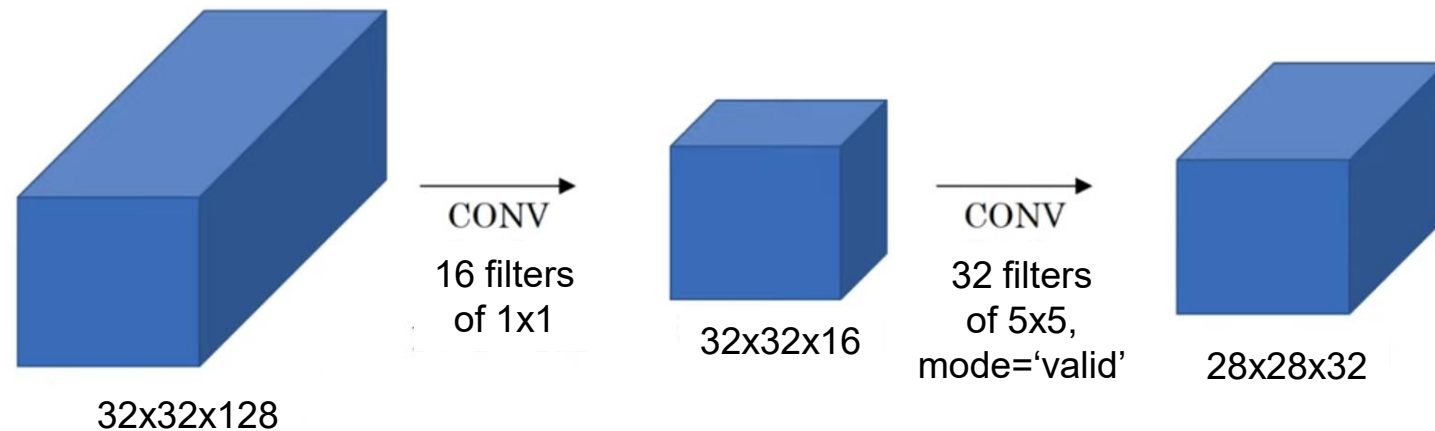
Question 29

- How many multiplications are performed in the figure below (where we move from $32 \times 32 \times 128$ to $28 \times 28 \times 32$, using 16 1×1 filters and 32 5×5 filters)?



Question 29

- How many multiplications are performed in the figure below (where we move from $32 \times 32 \times 128$ to $28 \times 28 \times 32$, using 16 1×1 filters and 32 5×5 filters)?



We have to perform:

a) $1 \times 1 \times 128 \times 32 \times 32 \times 16 = \sim 2.1\text{M mult.}$

b) $5 \times 5 \times 16 \times 28 \times 28 \times 32 = \sim 10\text{M mult.}$

Total (a+b): $\sim 12.1\text{M mult.}$

Question 30

- If you are working with a ConvNet whose training and validation accuracy are lower than expected, what would you think it would be a good strategy to follow?
 - A. Reduce the number of filters per convolutional block.
 - B. Increase regularization.
 - C. Reduce network depth.
 - D. None of the above.

Question 31

- If we want to reduce dimensionality in a ConvNet, what can we do?
 - A. Use a larger stride.
 - B. Use larger filters/kernels (no padding).
 - C. Use pooling.
 - D. All are correct.

Question 32

- Evaluate whether the following statements are true or false.
 - A. Gradient descent uses backpropagation. True/False
 - B. Backpropagation uses gradient descent. True/False
 - C. The deeper layers of a neural network are typically computing more complex features of the input than the earlier layers. True/False
 - D. The earlier layers of a neural network are typically computing more complex features of the input than the deeper layers. True/False

Question 32

- Evaluate whether the following statements are true or false.

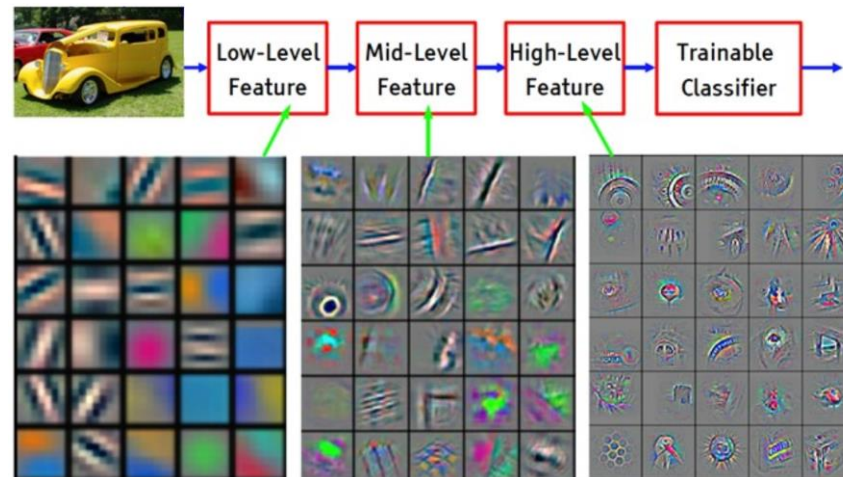
– Gradient descent uses backprop!

Optimization method for minimizing a loss function using the gradient

$$\mathbf{a}_{n+1} = \mathbf{a}_n - \gamma \nabla F(\mathbf{a}_n)$$

Method for calculating the gradient (i.e. the partial derivatives of the loss function with respect to the weights)

– The deeper layers of a neural network are typically computing more complex features of the input than the earlier layers



Question 33

- The following equations describe batch normalization:

$$\mu_j = \frac{1}{N} \sum_{i=1}^N x_{i,j} \quad \text{Per-channel mean} \qquad \sigma_j^2 = \frac{1}{N} \sum_{i=1}^N (x_{i,j} - \mu_j)^2 \quad \text{Per-channel var}$$

$$\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \varepsilon}} \quad \text{Normalized x} \qquad y_{i,j} = \gamma_j \hat{x}_{i,j} + \beta_j \quad \text{Output}$$

What would be the values for γ and β that the network should learn to recover the identity function ($y_{i,j} = x_{i,j}$)?

- A. $\gamma = 0$ y $\beta = 1$
- B. $\gamma = 1$ y $\beta = 0$
- C. $\gamma = \sigma$ y $\beta = \mu$
- D. $\gamma = \mu$ y $\beta = \sigma$

Question 34

- Fill the table with the most common configuration of loss function and output units activation function.

Problem	Loss Function	Output units activation function
Regression	(a)	(d)
Binary Classification	(b)	(e)
Multiclass Classification	(c)	(f)

- 1 – Sigmoid
- 2 – Mean Squared Error
- 3 – Binary Crossentropy
- 4 – Softmax
- 5 – Identity/Linear
- 6 – Categorical Crossentropy

Question 35

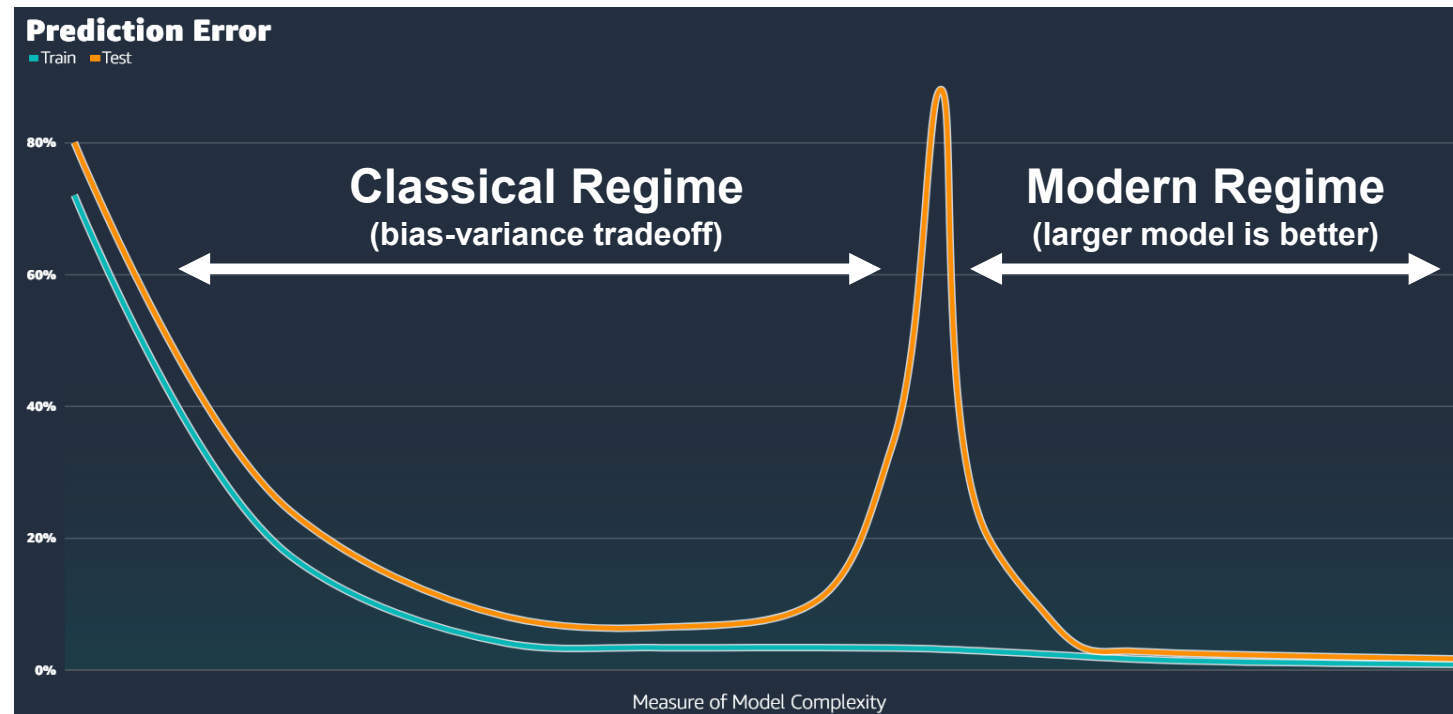
- Which of the following are reasonable ways of regularizing a neural network? Mark all that apply.
 - A. Introduce loss penalties
 - B. Reduce the learning rate
 - C. Introduce early stopping
 - D. Introduce data augmentation
 - E. Remove dropout
 - F. Reduce batch size

Question 36

- Which of the following is a characteristic of the phenomenon known as "deep double descent"?
 - A. The out-of-sample performance consistently improves as model complexity increases.
 - B. Overfitting increases as model complexity increases.
 - C. Double descent leads to the model out-of-sample performance first increasing and then decreasing as the number of training parameters increases, with a second increase at high model complexity.
 - D. Double descent leads to the model out-of-sample performance first decreasing and then increasing as the number of training parameters increases, with a second decrease at high model complexity.
 - E. Deep double descent only occurs when training data is very large and noisy.

Question 36

Deep double descent



Recommended readings:

<https://mlu-explain.github.io/double-descent/>
<https://openai.com/research/deep-double-descent>

Schaeffer, Rylan, et al. "Double Descent Demystified: Identifying, Interpreting & Ablating the Sources of a Deep Learning Puzzle." *arXiv preprint arXiv:2303.14151* (2023).

Question 37

- If we want to solve a multiclass classification problem, do we have to incorporate fully-connected (FC) layers at the end of our convolutional neural network?
 - A. Yes, we need, at least, one FC layer to classify our input images using softmax.
 - B. No, it's not mandatory to incorporate FC layers. We could use global average pooling or global max pooling, for instance.
 - C. No, it's not mandatory to incorporate FC layers. We could include instance normalization at the output.

Question 38

- When applying transfer learning in computer vision, the most commonly fine-tuned layers are:
 - A. Input layers
 - B. Layers near the output
 - C. Initial convolutional layers
 - D. Pooling layers

Question 39

- Which metrics are more appropriate for evaluating image regression models? Mark all that apply.
 - A. Accuracy
 - B. F1-score
 - C. Mean Average Precision (mAP)
 - D. Mean Absolute Error (MAE)

Question 40

- The primary purpose of batch normalization is to:
 - A. Reduce overfitting.
 - B. Accelerate training.
 - C. Enhance the interpretability of features.
 - D. Improve model initialization.

Question 40

- The primary purpose of batch normalization is to:
 - A. Reduce overfitting.
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Batch normalization: Accelerating deep network training by reducing internal covariate shift

S Ioffe, [C Szegedy](#) - International conference on machine ..., 2015 - proceedings.mlr.press

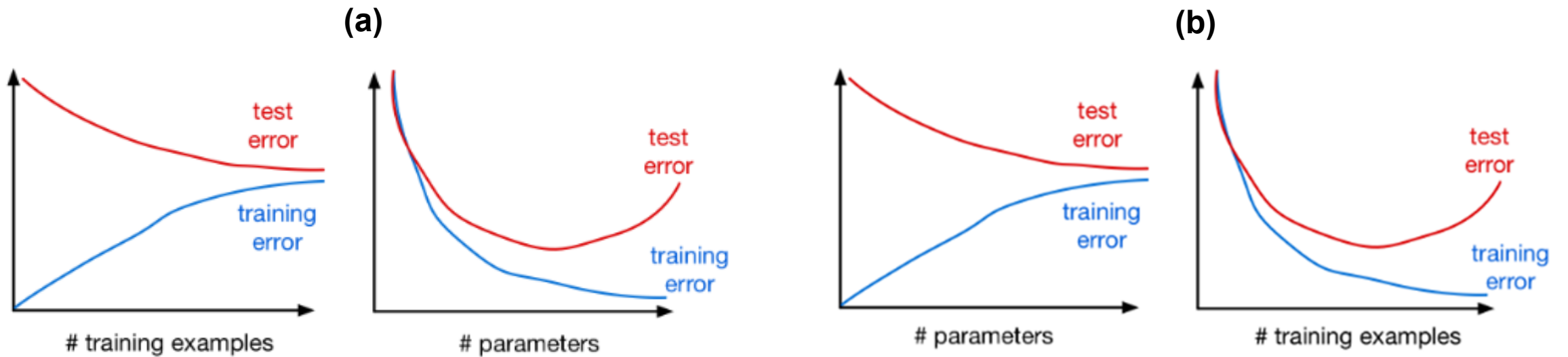
... 4.2, we apply **Batch Normalization** to the bestperforming ImageNet classification network, and

... Using an ensemble of such networks trained with **Batch Normalization**, we achieve the top-...

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Question 41

- Which of the following pairs of graphs is correct?



Question 42

- What is the main purpose of using dilated (atrous) convolutions in ConvNets?
 - A. To reduce the computational cost of convolutions without losing accuracy.
 - B. To increase the receptive field of a filter without increasing the number of parameters.
 - C. To downsample the feature maps during training.
 - D. To improve generalization by introducing sparsity in weights.

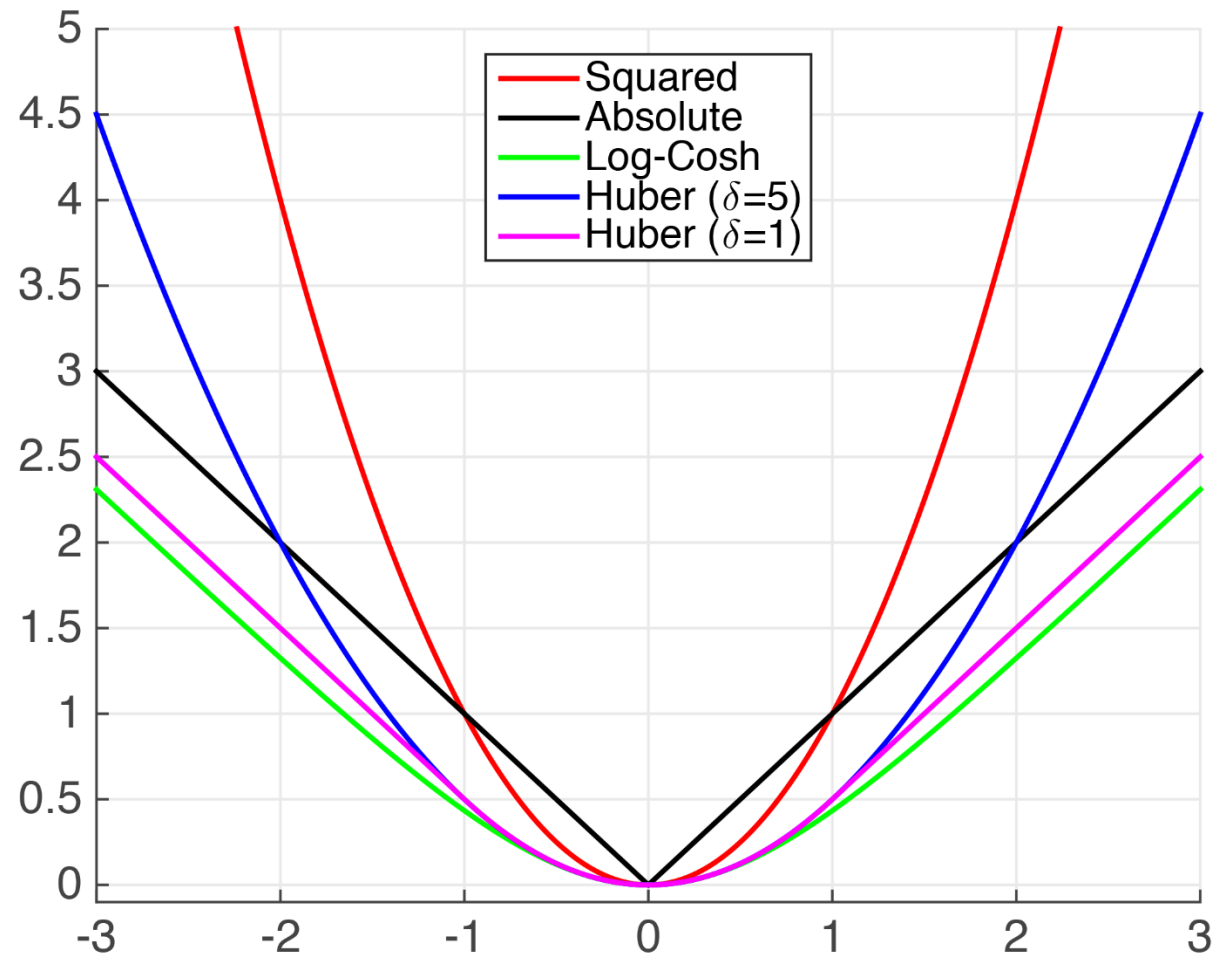
Question 43

- When adapting a ConvNet for regression, which of the following changes is typically necessary in the final layer?
 - A. Use a sigmoid activation function in the final layer.
 - B. Replace the softmax activation function with a linear activation function.
 - C. Reduce the depth of the network significantly.
 - D. Add a pooling layer before the final layer to aggregate features.

Question 44

- In image regression tasks using ConvNets, what is the impact of using mean absolute error (MAE) instead of mean squared error (MSE) as the loss function?
 - A. MAE penalizes larger errors more heavily than MSE.
 - B. MAE is less sensitive to outliers, so their impact on training is contained.
 - C. MAE converges faster but is less robust to outliers compared to MSE.
 - D. MAE is unsuitable for image regression tasks due to its non-linearity.

Question 44



See <https://myweb.uiowa.edu/pbreheny/uk/764/notes/12-1.pdf> or <https://arxiv.org/pdf/1808.09211>

Question 45

- In a ConvNet, why are larger kernels (e.g., 9×9) in the initial layers often replaced with smaller kernels (e.g., 3×3)?
 - A) Smaller kernels are more computationally efficient and allow to increase the depth of the ConvNet.
 - B) Smaller kernels reduce the depth of the ConvNet, improving generalization.
 - C) Smaller kernels can capture better features in the image.
 - D) Larger kernels are not compatible with deep learning frameworks.

Reviewing Deep Learning and Convolutional Neural Networks

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